Solution Requirements

Date	27 June 2026
Team ID	xxxxxx
Project Name	Emotion-Detection-Learning-Support-Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Current version does not require user login. Future versions may support secure user authentication for personalized experiences.	Low
2	Authorization Levels	The current system provides a single user role with access to all available features. Future versions may include Admin and User roles.	Low
3	External Interfaces	The system integrates with Wikipedia API for fact verification and Hugging Face Transformers for NLP-based event analysis.	High
4	Transaction Processing	The system processes user inputs (event description and interests), generates conversation suggestions, and stores conversation history and feedback.	High
5	Reporting	The system generates downloadable PDF reports containing detected topics and conversation starters.	Medium
6	Business Rules	Users must provide both event description and interests. The system analyzes event text, extracts topics, and generates personalized networking suggestions.	High
7	Compliance to Laws or Regulations	The application does not collect sensitive personal information. It follows general data privacy principles while storing user interactions locally.	Medium
8	Other	The system stores conversation history and user	Medium

S.No	Requirement Category	Requirement Description	Priority
		feedback for future reference and improvement.	

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate networking suggestions quickly and provide fast API responses.	Average response time < 2 seconds
2	Scalability	The system should support additional users and future cloud deployment.	Supports at least 100 concurrent users in future deployment
3	Security & Data Privacy	User data, feedback, and API interactions should be securely handled.	No sensitive user data stored; secure API communication
4	Reliability & Availability	The application should operate continuously without major failures.	System availability > 95% during testing
5	Usability & Accessibility	The interface should be intuitive, user-friendly, and easy to navigate.	Users can generate suggestions within 3 interaction steps
6	Maintainability & Portability	The code should be modular, reusable, and easy to deploy across different environments.	Modular architecture with reusable services and documented APIs